

Motor Datasheet

Operating Point: Continuous | Temperature: 25 °C

Geometry (active elements only)

Parameter	Value	Unit
Machine Outer Diameter	55.0	mm
Machine Inner Diameter	20.0	mm
Machine Thickness / Width	24.2	mm
Airgap	1.0	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	272.3	g
Rotor Mass	61.1	g
Stator Mass	211.2	g
Rotor Moment of Inertia (active elements only)	248.7	g cm2

Electrical Properties

Parameter	Value	Unit
Supply Current ('RMS')	3.9	A
Phase Resistance	86.0	mOhm
D-Axis Inductance (Ld)	255.3	μH
Q-Axis Inductance (Lq)	253.7	μH
PM Flux Linkage	2.7	mWb
Frequency at operating point (electrical)	237.5	Hz
Winding Configuration	wye (star)	-

Performance

Parameter	Value	Unit
Torque (Continuous)	111.0	mNm
Speed	2850.0	rpm
BackEMF Constant (Ke)	1.5	mV/rpm
Motor Constant (Km, only resistive losses)	53.5	mNm/sqrt(W)
Specific Motor Constant (Km / mass)	196.6	mNm/sqrt(W)/kg
Torque Constant (Kt, no-load)	19.9	mNm/A_peak
Torque Ripple	1.4	%
Saturation	3.5	%

Losses

Parameter	Value	Unit
Total Losses	4.6	W
Resistive Losses	3.9	W

Configuration

Parameter	Value	Unit
Number of Stators	1	-
Number of Rotors	1	-
Number of Slots	12	-
Pole Pairs	5	-
Number of Parallel Paths	1	-
Number of Turns per Tooth (designed incl. parallel paths)	22	-
Number of Phase Turns (all phases in series)	88	-

Materials

Component	Material	
Stator Core	700HR 5P 800MPa-Hoganas	-
Magnets	BMHFa-32/27	-

3D Model

